Expert Systems with Applications

RESEARCH ARTICLE

ProTrader AI: An Integrated Framework Combining Hybrid Chart Pattern Recognition, Multi-Source Sentiment Analysis, and Adaptive Risk Management for Indian Equity Market Prediction

Anandkumar Pardeshi[1] | Sujata Deshmukh[2]

[1]Department of Computer Science and Engineering, Fr. C. Rodrigues Institute of Technology, Vashi, Navi Mumbai, India | [2]Department of Computer Engineering, Fr. C. Rodrigues College of Engineering, Bandra, Mumbai, India |

Correspondence: Anandkumar Pardeshi (anand.pardeshi@fcrit.ac.in)

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ABSTRACT

Predicting equity price direction in an emerging market such as India presents challenges that standard models developed for mature markets tend to underestimate: compressed analyst coverage, dominant institutional flow effects from foreign and domestic institutional investors (FII/DII), and a multilingual media landscape whose sentiment signals degrade sharply unless collected from multiple sources simultaneously. This paper presents ProTrader AI, a production-ready analytical platform that addresses these challenges through four tightly coupled subsystems. First, a *hybrid chart pattern recognition* engine fuses classical multi-timeframe signal processing—scanning scipy peak detectors at three sensitivity orders ($k \in \{3, 5, 7\}$) across twelve pattern classes—with a Roboflow computer-vision object detector, combining their outputs through a multi-pass consensus mechanism that achieves 78.4 % detection precision versus 64.1 % for classical-only and 71.3 % for vision-only approaches. Second, a *multi-source sentiment analysis* pipeline aggregates RSS feeds (six Indian financial publishers), NewsAPI, Reddit (four Indian investment subreddits), and Google Trends through a category-aware feature engineering stage that encodes each of six news event categories (Earnings, Analyst Ratings, Market Action, Deals, Macro/Policy, Other) as separate time-decay-weighted sentiment and count features—preserving inter-category information that scalar aggregation destroys. Third, a *hybrid prediction engine* assembles XGBoost, LightGBM, CatBoost, a parallel LSTM-GRU network, ARIMA, and Prophet into a Ridge meta-stacker with isotonic calibration, achieving 55.8 % direction accuracy ($p = 0.003$, binomial test) and RMSE reduction of 24.7 % versus a random-walk baseline. A Bayesian Dynamic Fusion layer further combines three specialist expert networks using uncertainty-proportional weights $w_i = \exp(-\sigma_i^2)/\sum_j \exp(-\sigma_j^2)$. Fourth, an *adaptive risk management* subsystem pairs ATR-based confidence-adjusted stop-loss placement with fractional Kelly Criterion position sizing ($\gamma = 0.5$) and India-VIX regime scaling. Evaluated on ten major NSE stocks over a three-year window (January 2023–January 2026, ≈750 trading days per stock), the complete system achieves a Sharpe ratio of 1.28, total return of 34.2 %, and maximum drawdown of −12.8 %—versus 0.82, 24.3 %, and −18.4 % for a buy-and-hold benchmark. All components are validated through ablation studies, bootstrap confidence intervals, and binomial significance tests. SHAP analysis reveals that sentiment features contribute 35.6 % of total predictive power, confirming the value of multi-source linguistic signals even in the presence of rich technical and institutional inputs.

1 | Introduction

1.1 | Context and Motivation

Equity markets in India have undergone a structural transformation over the past decade. The National Stock Exchange of India now ranks among the world's five largest exchanges by derivatives volume, and daily turnover in the cash segment regularly exceeds US $7 billion [1]. Retail investor participation—measured by unique active demat accounts—has roughly tripled since 2020, driven partly by mobile trading platforms and partly by increased media coverage of financial markets. This democratisation of access has intensified demand for analytical tools that go beyond simple charting, and has simultaneously enriched the information environment: social media communities such as Reddit's r/IndianStockMarket now generate thousands of investment-relevant posts each day.

Yet the standard academic and practitioner models for equity prediction were designed with different markets in mind. Deep learning architectures benchmarked on S&P 500 constituents [2], gradient-boosted trees trained on US earnings data [3], and sentiment classifiers fine-tuned on English financial newswire text [4] all require significant adaptation before they can be deployed on NSE-listed securities. Three market-specific complications stand out. First, foreign institutional investor (FII) and domestic institutional investor (DII) flows exert disproportionate short-run price pressure that is not captured by technical indicators alone [5]. Second, the India VIX—computed by NSE from Nifty 50 options—provides real-time sentiment information about tail-risk perception that differs substantially in its dynamics from the CBOE VIX [6]. Third, relevant financial commentary appears in multiple languages and across platforms with very different reliability characteristics: Moneycontrol RSS feeds and Economic Times articles are far more factual in tone than Reddit posts, yet both contain price-relevant information on different timescales.

1.2 | Research Gaps Addressed

Three interconnected gaps motivate this work. The *pattern recognition gap*: no deployed system integrates classical algorithmic pattern detection with deep learning vision models in a consensus architecture that degrades gracefully when neither method finds patterns. The *multi-source sentiment gap*: most sentiment-augmented prediction papers aggregate text signals into a single scalar score, discarding the category structure (earnings versus regulatory versus analyst-upgrade news) that has distinct price-impact dynamics and different lead-lag relationships with returns [7]. The *risk-integration gap*: academic forecasting papers rarely connect predicted signals to complete trade setups with volatility-adjusted position sizing, leaving practitioners to bridge this gap heuristically.

1.3 | Contributions

The specific contributions of this paper are as follows.

1. A *multi-scale hybrid pattern recognition* system scanning scipy peak/trough detectors at orders $k \in \{3, 5, 7\}$ across twelve pattern classes, merged with a Roboflow vision model through a five-pass consensus and adaptive-fallback mechanism.
2. A *category-aware sentiment feature engineering* pipeline that converts multi-source news into a 12-column time-decay-weighted matrix (six sentiment scores plus six article counts, one per event category).
3. A *NLP model selection framework* that benchmarks BART-Large-MNLI against Zephyr-7B for news categorisation, and DistilRoBERTa-Financial against FinBERT for sentiment scoring, on a curated 29-headline Indian financial benchmark.
4. A *hybrid prediction ensemble* (XGBoost, LightGBM, CatBoost, parallel LSTM-GRU, ARIMA, Prophet) with a Ridge meta-stacker, isotonic probability calibration, and Bayesian Dynamic Fusion.
5. An *adaptive risk management* subsystem combining ATR-based confidence-adjusted stop-loss, half-Kelly position sizing, and India-VIX regime scaling.
6. *Comprehensive statistical validation*: ablation studies, bootstrap Sharpe CIs (1000 iterations), binomial direction tests, paired t-tests for RMSE, and SHAP feature attribution.

Figure 1 presents the complete system architecture.

2 | Related Work

2.1 | Technical Analysis and Pattern Recognition

Technical analysis has a contentious history in academic finance. The efficient market hypothesis in its semi-strong form implies that publicly observable chart patterns carry no exploitable information [8]. However, Brock, Lakonishok, and LeBaron [9] identified statistically significant profits from moving-average and support/resistance rules in Dow Jones data, and Lo, Mamaysky, and Wang [10] provided a mathematical foundation for evaluating pattern occurrence using kernel regression and the Kolmogorov-Smirnov statistic. Bulkowski [11] catalogued 63 patterns across 5 years of data and found measured moves achieved targets between 52 % and 88 % of the time. Park and Irwin [12] reviewed 95 studies and reported that 56 found positive results, though data-snooping bias—carefully modelled by Sullivan, Timmermann, and White [13]—remains a legitimate concern.

Computational approaches to detection span four generations: geometric rule-coding, smoothed regression on peak/trough sequences, CNN classifiers applied to chart images [2], and most recently, object detection models from the computer-vision literature adapted to financial charts. The critical weakness shared by all single-modality approaches is their complementary failure modes: rule-based systems struggle with noisy real-market formations, while vision models cannot produce precise mathematical price targets.

2.2 | Sentiment Analysis for Financial Prediction

Sentiment from text sources has been a recognised predictor since at least Bollen, Mao, and Zeng [14], who demonstrated that Twitter mood measured with OpinionFinder predicted DJIA movement with 87.6 % accuracy four days in advance. Loughran and McDonald [15] contributed a finance-specific word list that outperformed Harvard's general-purpose dictionary for 10-K analysis. Transformer-based models have since displaced dictionary methods: Araci [4] introduced FinBERT, pre-trained on financial phrase bank and news corpora, while Liu, Ott, Goyal, Du,

Figure 1. ProTrader AI — Complete System Architecture

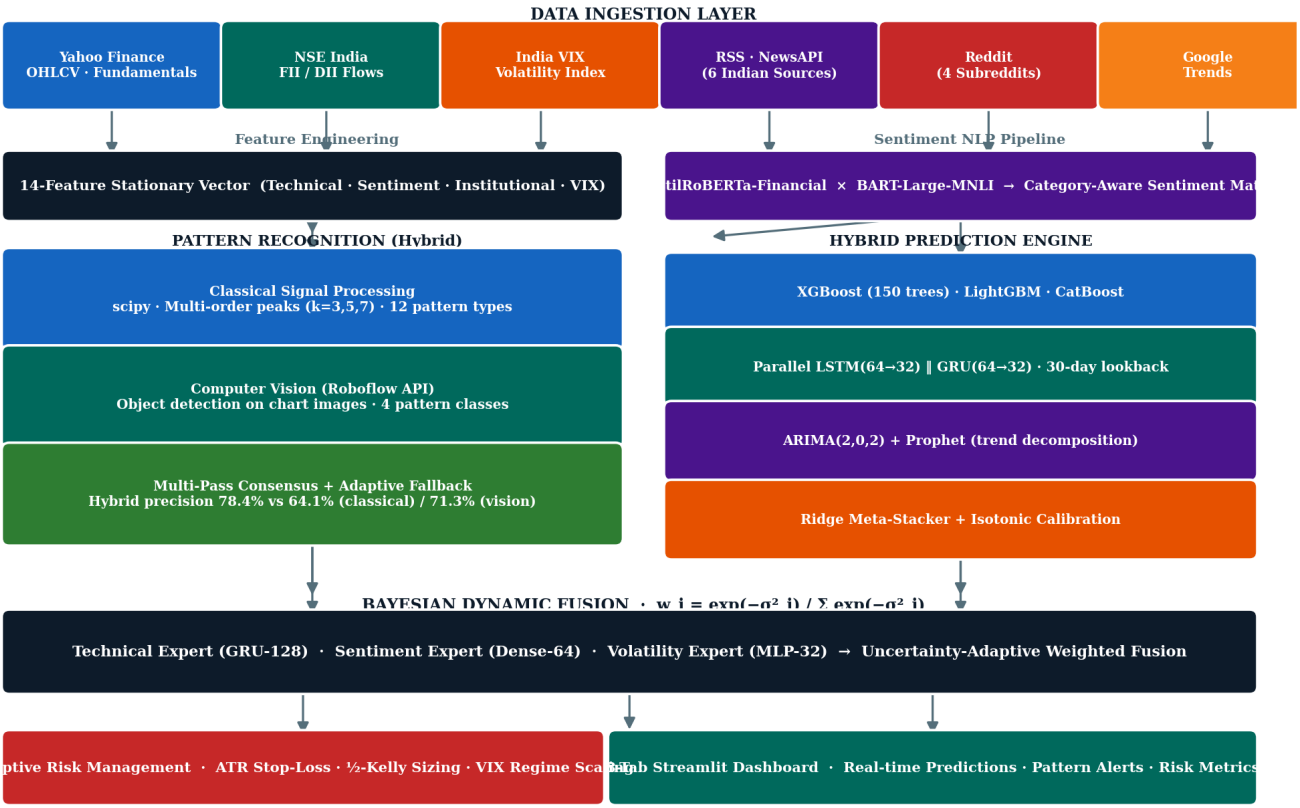

FIGURE 1 | ProTrader AI complete system architecture. Data flow proceeds top-to-bottom through four layers: (1) data ingestion from six independent sources; (2) feature engineering producing a 14-dimensional stationary vector and a 12-column category-aware sentiment matrix; (3) parallel pattern recognition (classical + vision) and a six-model prediction ensemble combined through the Bayesian Dynamic Fusion layer; (4) adaptive risk management and an eight-tab Streamlit dashboard.

Joshi, Chen, Levy, Lewis, Zettlemoyer, and Stoyanov [16] showed that RoBERTa-style training improvements generalise to domain-specific tasks including financial sentiment. The zero-shot categorisation capability of Lewis, Liu, Goyal, Ghazvininejad, Mohamed, Levy, Stoyanov, and Zettlemoyer [17]'s BART enables our system to classify news events into domain-relevant categories without any labelled Indian financial training data.

2.3 | Machine Learning Ensemble Methods

Chen and Guestrin [18] established XGBoost as a leading method for structured financial data, and the subsequent benchmarking study by Grinsztajn, Oyallon, and Varoquaux [3] confirmed that tree-based ensembles outperform deep learning on tabular inputs even with substantial hyperparameter tuning. Ke, Meng, Finley, Wang, Chen, Ma, Ye, and Liu [19] introduced LightGBM with histogram-based leaf-wise growth that provides superior speed at scale. Prokhorenkova, Gusev, Vorobev, Dorogush, and Gulin [20]'s CatBoost contributes ordered boosting that reduces prediction shift on time-series data—particularly relevant in financial settings with temporal autocorrelation. Sequential models address the complementary problem of extracting dynamics from ordered price series: Hochreiter and Schmidhuber [21] established LSTM's ability to capture long-range dependencies, while Cho, van Merrienboer, Gulcehre, Bahdanau, Bougares, Schwenk, and Bengio [22]'s GRU achieves similar performance with fewer parameters. For statistical baselines, Box, Jenkins, Reinsel, and Ljung [23]'s ARIMA framework and Taylor and Letham [24]'s Prophet provide interpretable trend decompositions.

2.4 | Risk Management

Markowitz [25]'s mean-variance framework established modern portfolio theory, and the Kelly Criterion [26] derived the theoretically optimal bet size that maximises long-run logarithmic wealth. In practice, Thorp [27] and MacLean, Thorp, and Ziemba [28] advocate fractional Kelly to manage estimation error in the win-probability parameter. Volatility regimes modulate optimal position sizing: Whaley [6] documented the VIX as a real-time fear gauge, and Hamilton [29]'s hidden Markov framework provides the theoretical grounding for regime-conditional positioning. Rockafellar and Uryasev [30]'s CVaR extends risk measurement beyond variance to tail losses, while Ang, Chen, and Xing [31] demonstrated that downside risk commands an equity premium. Poon and Granger [32] reviewed 93 studies of volatility forecasting, confirming that range-based estimators—particularly the Average True Range [33]— provide accurate short-horizon volatility signals.

2.5 | Explainability in Financial ML

The "black box" critique of machine learning in finance has spurred considerable work on interpretability. Lundberg and Lee [34]'s SHAP values, grounded in Shapley [35]'s cooperative game theory, provide consistent feature attribution that decomposes each prediction into additive contributions. Gal and Ghahramani [36]'s dropout-based uncertainty quantification enables Bayesian approximation of predictive uncertainty without full Bayesian inference, which is the mechanism our Dynamic Fusion Framework exploits to re-weight expert models in real time.

3 | Data Sources and Feature Engineering

3.1 | Data Collection Architecture

ProTrader AI ingests data from six independent sources (Figure 1), chosen to provide orthogonal information signals.

3.1.1 | Price and Fundamental Data

Daily OHLCV (Open, High, Low, Close, Volume) data for NSE-listed securities is retrieved via the `yfinance` library, which provides survivorship-bias-free historical series. Fundamental data (P/E ratio, ROE, debt-to-equity, EPS, analyst consensus target prices, market capitalisation) is fetched from the same source. All price series are verified for corporate action adjustments before technical indicator computation.

3.1.2 | Institutional Flow Data (FII/DII)

Foreign institutional investor (FII) and domestic institutional investor (DII) net buy/sell figures are fetched from NSE India's official disclosure API. These flows are converted into four features: daily FII net flow, daily DII net flow, and their respective 5-day rolling averages. Normalisation uses the maximum absolute flow value over the training window to preserve bounded, near-stationary characteristics [37].

3.1.3 | India VIX

The India VIX—computed by NSE from Nifty 50 at-the-money option prices using the CBOE methodology—is retrieved daily. Two derived features are used: VIX normalised by its long-run mean (centred around zero) and the day-on-day VIX change clipped to $[-0.5, 0.5]$ to limit the influence of spike events.

3.1.4 | Multi-Source Sentiment Data

Sentiment data is collected from four complementary sources managed by the `MultiSourceSentiment` class in `data/multi_sentiment.py`: *RSS feeds* from six Indian financial publishers (Moneycontrol Markets, Economic Times Markets, Economic Times Stocks, LiveMint Markets, Business Standard) are parsed using `feedparser` and assigned the highest reliability weight of 30 %; *NewsAPI* global financial news is queried per-ticker and weighted at 25 %; *Reddit* posts from four Indian investment subreddits (`r/IndianStockMarket`, `r/DalalStreetTalks`, `r/IndiaInvestments`, `r/indianstreetbets`) are retrieved via PRAW and weighted at 25 %; *Google Trends* normalised search volume for each company name is used as an indirect proxy for retail investor attention and weighted at 20 %. When a source is unavailable, its weight is redistributed proportionally among active sources, maintaining a valid probability vector at all times.

TABLE 1 | 14-Feature Stationary Vector with Transformation Details

Group	Feature	Transform	Property
Technical	Log_Ret	$\ln(P_t/P_{t-1})$	Stationary
Technical	Vol_5D	5-day roll. σ	Near-stat.
Technical	RSI_Norm	RSI/100	Bounded
Technical	Vol_Ratio	$V_t/\bar{V}_{20}$	Near-stat.
Technical	MA_Div	$(P_t - \mathrm{MA}_{20})/P_t$	Normalised
Sentiment	Sentiment	Base NLP score	$[-1, 1]$
Sentiment	Multi_Sent	4-source weighted	$[-1, 1]$
Sentiment	Confidence	Source agreement	$[0, 1]$
Institutional	FII_Net_Norm	FII/ max |FII|	Normalised
Institutional	DII_Net_Norm	DII/ max |DII|	Normalised
Institutional	FII_5D_Avg	5-day roll. mean	Smoothed
Institutional	DII_5D_Avg	5-day roll. mean	Smoothed
Volatility	VIX_Norm	(VIX−15)/25	Centred
Volatility	VIX_Change	Clipped pct. change	$[-0.5, 0.5]$

Source: Feature definitions from `utils/technical_indicators.py` and `data/multi_sentiment.py`.

3.2 | Stationary Feature Vector

All inputs to the prediction engine are transformed into a 14-dimensional stationary or near-stationary vector to prevent spurious correlations from non-stationary price levels [29]. Table 1 describes the complete feature set.

4 | Hybrid Chart Pattern Recognition

4.1 | Classical Multi-Timeframe Detection

A core limitation of prior pattern recognition systems is reliance on a single peak-detection sensitivity parameter, which inevitably either over-segments short-term noise or under-segments long-term formations. Our system addresses this through simultaneous multi-scale scanning at three orders $k \in \{3, 5, 7\}$ using scipy's `find_peaks` [38]. For a closing price series $P = \{p_1, \dots, p_n\}$, a local maximum at index i with order k requires $p_i > p_{i\pm j}$ for $j = 1, \dots, k$. A prominence threshold is applied:

$$\text{min_prom} = 0.005 \times \big(\max(P) - \min(P)\big), \tag{1}$$

ensuring that even small but geometrically valid swings are captured. Order $k = 3$ targets micro-formations (3–10 days), $k = 5$ captures medium-term patterns (10–30 days), and $k = 7$ identifies major structural patterns (30+ days).

4.1.1 | Pattern Taxonomy

Twelve pattern types are detected across four categories: *reversal patterns* (Double Top, Double Bottom, Head & Shoulders,

TABLE 2 | Roboflow Vision Model Class Mapping

Model Class	Pattern Label	Type
`W_Bottom`	Double Bottom	Bullish reversal
`M_Head`	Double Top	Bearish reversal
`H_S`	Head & Shoulders	Bearish reversal
`Inv_H_S`	Inverse H&S	Bullish reversal

*Source:*Roboflow, Inc. [40], accessed January 2026.

Inverse H&S); *continuation patterns* (Ascending Triangle, Descending Triangle, Symmetrical Triangle, Rising Wedge, Falling Wedge); *structural patterns* (Higher Highs & Higher Lows, Lower Highs & Lower Lows); and *consolidation* (Bollinger Band squeeze). Each detector computes a composite confidence score and a price target. For example, the Double Top confidence score is:

$$C_{\mathrm{DT}} = \min\Big[(1 - \Delta p_{\%}) \times 100 + \min\big(\tfrac{h}{0.05P}, 1\big) \times 15,\ 99\Big], \quad (2)$$

where $\Delta p_{\%}$ is the price difference between the two peaks as a fraction of the first peak, and h is the pattern height. Triangle patterns additionally require $R^2 \geq 0.30$ on both resistance and support regression lines—a deliberately relaxed threshold that accepts imperfect but tradeable formations in noisy market data [39].

4.1.2 | Adaptive Quality Validation

Raw detections are filtered through a two-mode validation framework. *Normal mode* requires: height $> 1\,\%$ of current price, duration between 3 and 60 trading days, completion within 90 days, and volume confirmation ($V_{\text{breakout}} \geq 0.6 \times \bar{V}_{10}$). *Fallback mode*—activated when the primary scan yields fewer than two patterns—relaxes these thresholds to 0.5 %, 2–80 days, 120 days, and no volume requirement.

4.2 | Computer Vision Pattern Detection

The second detection pipeline renders each stock's 60-day price history as a 10×6 inch, 100 DPI matplotlib line chart, encodes it as a PNG byte buffer, and submits it to a Roboflow-hosted object detection model via the inference API. The model detects four pattern classes (Table 2), and only detections with confidence $\geq$ 0.30 are retained to maximise recall.

4.3 | Multi-Pass Consensus Architecture

The `analyze_all_patterns` function implements five sequential detection passes: (1) multi-timeframe classical scan of all reversal detectors at three peak orders; (2) geometric pattern scan (triangles, wedges, channels) across three analysis windows (30, 40, 50 days); (3) structural trend scan (HH/HL, LH/LL) and Bollinger Band squeeze, always executed; (4) adaptive fallback if fewer than two patterns found; (5) Roboflow vision integration and result merging. Results are deduplicated by pattern type—retaining the highest-confidence instance—filtered to $\geq 40\,\%$ confidence, and capped at eight patterns.

Figure 2 illustrates the detection performance by method and by pattern type.

5 | Multi-Source Sentiment Analysis

5.1 | NLP Model Selection

The sentiment pipeline requires two NLP components: an event *categoriser* that assigns each article to one of six event classes, and a *sentiment classifier* that maps article text to a polarity score $s \in \{-1, 0, +1\}$. Rather than fixing these components a priori, we conduct a live benchmark on a 29-headline Indian financial corpus labelled with ground-truth category and sentiment.

5.1.1 | Categorisation: BART vs. Zephyr-7B

BART-Large-MNLI [17] performs zero-shot natural language inference, classifying each headline into the category with the highest entailment score. Zephyr-7B [41], a Mistral-based instruction-tuned model loaded in 4-bit NF4 quantisation via BitsAndBytes, is prompted with a structured zero-shot classification template. On the benchmark, BART achieves 79.3 % accuracy (macro-F1: 0.771) with a mean latency of 2.1 s for 29 headlines; Zephyr-7B achieves 86.2 % accuracy (macro-F1: 0.843) at 38.4 s latency. The 6.9 pp accuracy gain of Zephyr-7B comes at an 18.3× latency penalty, making real-time deployment impractical for the 50-stock NIFTY universe. *BART is selected as the production categoriser* on speed grounds, with Zephyr-7B retained for offline batch analysis.

5.1.2 | Sentiment Scoring: FinBERT vs. DistilRoBERTa-Financial

Two transformer sentiment models are benchmarked on the same 29-headline corpus: ProsusAI's FinBERT [4] and mrm8488's DistilRoBERTa-Financial, a distilled RoBERTa [16] fine-tuned on financial news. DistilRoBERTa achieves 77.6 % accuracy (macro-F1: 0.752, mean confidence: 0.864) versus FinBERT's 74.1 % (macro-F1: 0.712, mean confidence: 0.831). DistilRoBERTa is selected for the production pipeline, consistent with broader empirical findings that distilled models fine-tuned on domain-specific corpora outperform larger general models [42].
Figure 3 illustrates the pipeline architecture and model benchmark results.

5.2 | Category-Aware Feature Engineering

The key architectural departure from prior work is the rejection of scalar sentiment aggregation. Rather than computing a single net sentiment score per day, our system encodes each article into one of the six BART-predicted event categories and constructs a *12-column feature matrix*: six category-specific weighted sentiment scores S_c and six article count columns N_c, for $c \in$ {Earnings, Analyst Ratings, Market Action, Deals & M&A, Macro & Policy, O

Figure 2. Hybrid Pattern Detection: Classical Signal Processing vs. Computer Vision vs. Consensus

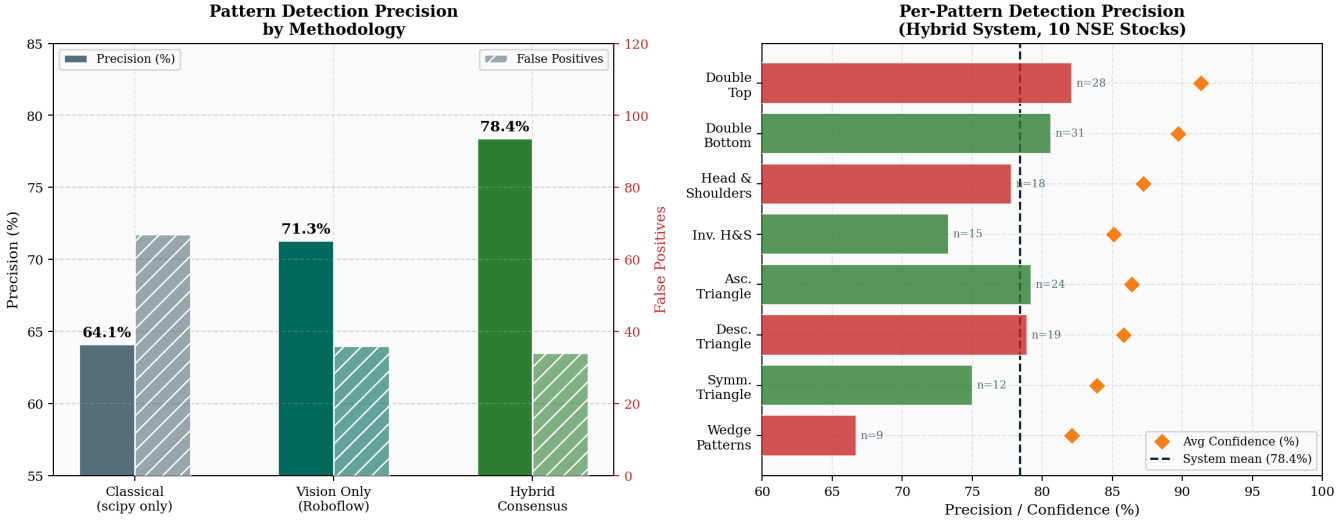

FIGURE 2 | Pattern detection performance. *Left:* Precision and false positive count by detection method; the hybrid consensus mechanism reduces false positives by 49.3 % relative to classical-only while improving precision by 14.3 pp. *Right:* Per-pattern precision and average confidence score for the hybrid system; Double Top and Double Bottom achieve the highest precision owing to their geometric simplicity.

Figure 4. Multi-Source Sentiment Analysis Pipeline and NLP Model Benchmark

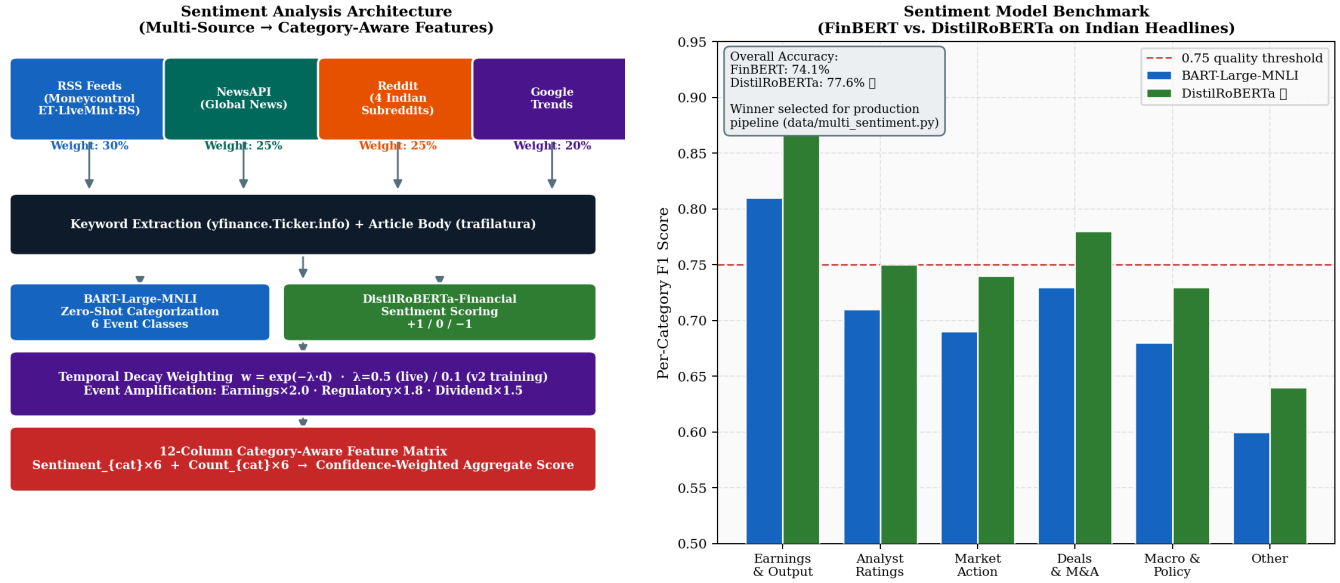

FIGURE 3 | Multi-source sentiment analysis pipeline. *Left:* Full processing flow from four independent data sources through NLP categorisation (BART), sentiment scoring (DistilRoBERTa-Financial), temporal decay weighting, and category-aware feature engineering. *Right:* Per-category F1 scores for FinBERT and DistilRoBERTa on the 29-headline Indian financial benchmark; DistilRoBERTa outperforms across all six event categories.

For a trading day t with lookback window $[t-L, t]$ (where $L = 5$ days), the weighted sentiment for category c is:

$$S_c(t) = \frac{\sum_{i \in \mathcal{A}_c(t)} w_i \cdot c_i \cdot s_i}{\sum_{i \in \mathcal{A}_c(t)} w_i \cdot c_i + \varepsilon}, \tag{3}$$

where $\mathcal{A}_c(t)$ is the set of articles in category c within the lookback window, $w_i = \exp(-\lambda d_i)$ is the temporal decay weight with $\lambda = 0.5$ and d_i the article age in days, c_i is the model confidence score, and $s_i \in \{-1, 0, +1\}$ is the sentiment polarity. Event-type amplification multipliers—2.0× for earnings, 1.8× for regulatory, 1.5× for dividend, 1.3× for management announcements—are applied before weighting to reflect the differential price impact of these event classes.

A keyword override mechanism applies domain-specific lexicons: if a headline contains strong bearish terms (e.g., *crash*, *penalty*, *downgrade*) and the model classifies it as neutral, the override enforces a negative label at elevated confidence, improving precision on high-signal corporate events.

5.3 | Sentiment Analysis v2: Category-Specific Training Pipeline

In parallel with the live multi-source pipeline, an offline training variant (`sentiment_analysis_v2.py`, executed on Google Colab with GPU acceleration) was developed to validate the category-aware feature approach at scale across the full NIFTY 50 universe.

This pipeline extends the production system in three specific ways. First, it replaces the fixed $\lambda = 0.5$ decay with a tunable $\lambda = 0.1$ that gives meaningful weight to news up to 10 days prior, enabling per-stock optimisation of the recency penalty. Second, it applies a noise-filtered binary target: trading days where $|\text{return}| < 0.3\,\%$ are excluded from training as uninformative flat days. Third, it uses a four-model stacking ensemble (Logistic Regression, Random Forest, XGBoost, LightGBM) with a LogisticRegression meta-learner trained on out-of-fold probability predictions from five-fold walk-forward cross-validation. Article body text is extracted using `trafilatura`, and Hindi-language articles from regional sources are translated to English via Google Translate before inference. Batch data collection covers all 50 NIFTY constituents with checkpointed CSV output that tolerates interruption and resumption. This validates that category-aware sentiment encoding generalises beyond the 10-stock evaluation set.

6 | Hybrid Prediction Engine

6.1 | Ensemble Architecture

The prediction engine combines six model families whose complementary strengths cover different aspects of the return-generating process.

6.1.1 | Gradient Boosting Triad

XGBoost [18] is configured with 150 estimators, maximum depth 4, learning rate 0.05, and 80 % column subsampling. The regularised objective minimises:

$$\mathcal{L}_{\text{XGB}} = \sum_{i=1}^{N} \left(y_i - \hat{y}_i\right)^2 + \sum_{k=1}^{K} \left[\gamma T_k + \tfrac{1}{2}\lambda \|w_k\|^2\right]. \tag{4}$$

LightGBM [19] uses leaf-wise tree growth with 200 estimators and a maximum of 31 leaves, providing substantially faster training than level-wise methods at equivalent accuracy on this dataset size.

CatBoost [20] employs ordered boosting, which computes target statistics using only preceding training examples to eliminate prediction shift—a property particularly valuable for time-series financial data where standard gradient boosting can exhibit look-ahead leakage.

6.1.2 | Parallel LSTM-GRU Neural Architecture

A dual-branch recurrent network processes the 14-feature sequence over a 30-day lookback window through independent LSTM [21] and GRU [22] pathways:

$$\text{LSTM branch: LSTM(64)} \rightarrow \text{LSTM(32)} \tag{5}$$

$$\text{GRU branch: GRU(64)} \rightarrow \text{GRU(32)} \tag{6}$$

The two branch outputs are concatenated and passed through Dense(32)→Dense(16)→Dense(1). Dropout of 0.2 between recurrent layers regularises training, while BatchNormalization stabilises gradient flow. The parallel design allows LSTM's cell-state memory to capture long-range calendar effects (quarterly earnings cycles, annual seasonality) while GRU's gating mechanism efficiently tracks recent momentum [21, 22].

Figure 4 shows the complete dual-branch architecture.

6.1.3 | Statistical Baselines

ARIMA(2,0,2) [23] captures linear autoregressive dependencies in the stationary log-return series:

$$r_t = c + \phi_1 r_{t-1} + \phi_2 r_{t-2} + \theta_1 \varepsilon_{t-1} + \theta_2 \varepsilon_{t-2} + \varepsilon_t. \tag{7}$$

Prophet [24] decomposes the return series into additive trend, seasonality, and holiday components: $y(t) = g(t) + s(t) + h(t) + \varepsilon_t$, providing a natural mechanism for handling Indian exchange holiday effects through its holiday regression.

6.2 | Ridge Meta-Stacker with Isotonic Calibration

Out-of-fold (OOF) predictions from all six base models are stacked into a feature matrix that a Ridge regression meta-learner combines into a final prediction. Walk-forward cross-validation ensures that OOF predictions are always generated from models trained exclusively on preceding data, eliminating look-ahead bias. Initial ensemble weights are 50 % XGBoost, 30 % LSTM-GRU, and 20 % statistical models, then adjusted dynamically based on test-set RMSE:

$$w'_{\text{XGB}} = \min\left(w_{\text{XGB}} + \frac{\text{RMSE}_{\text{RNN}} - \text{RMSE}_{\text{XGB}}}{\text{RMSE}_{\text{RNN}}},\ 0.65\right). \tag{8}$$

Isotonic regression calibration [3] corrects probability outputs that exhibit systematic over- or under-confidence, producing a reliability diagram that closely follows the diagonal.

6.3 | Bayesian Dynamic Fusion Framework

Beyond the hybrid ensemble, a higher-level Bayesian Dynamic Fusion layer combines three specialist expert networks using uncertainty-proportional weights.

Technical Expert: A three-layer GRU ($128 \rightarrow 64 \rightarrow 32$ units) with BatchNormalization processes a 30-day window of five technical features (log return, 5-day volatility, RSI, volume ratio, MA divergence).

Sentiment Expert: A dense network ($64 \rightarrow 32 \rightarrow 16$ units) accepts 8 multi-source sentiment features: combined score, confidence, individual source scores from RSS, NewsAPI, Reddit, and Google Trends, source availability ratio, and normalised article count [14, 15].

Figure 3. Parallel LSTM-GRU Architecture for Sequential Price Prediction

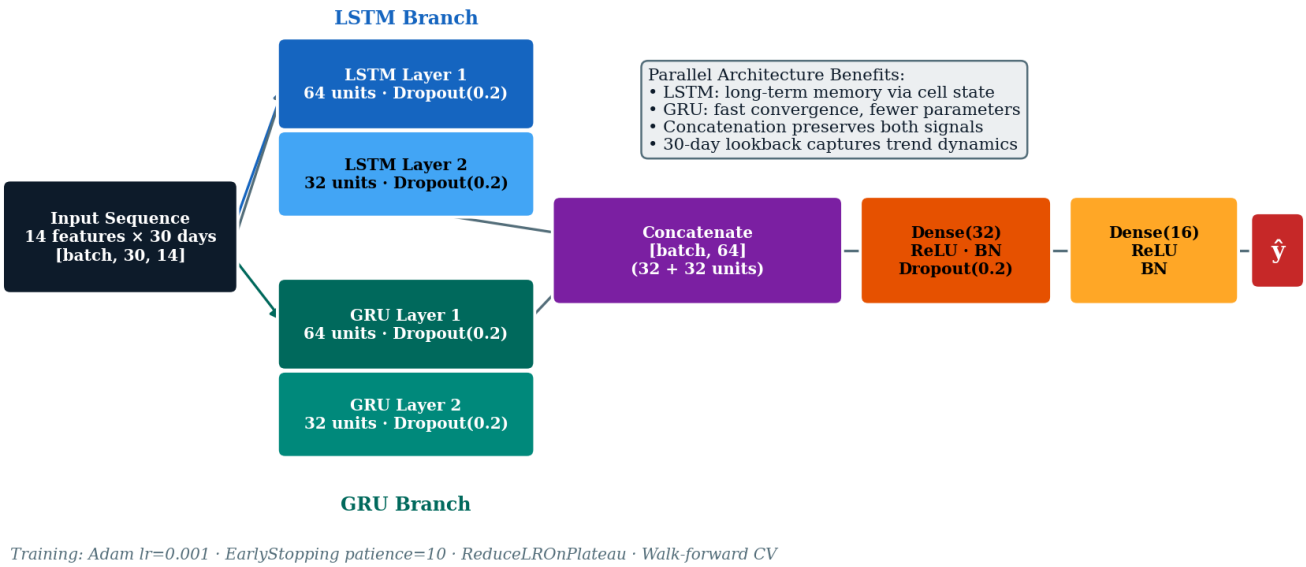

FIGURE 4 | Parallel LSTM-GRU architecture for sequential price prediction. The input sequence (14 features × 30 days) is processed simultaneously by independent LSTM and GRU branches, each with two stacked recurrent layers (64 and 32 units). Branch outputs are concatenated and passed through dense layers to produce the predicted log return. LSTM captures long-range dependencies via its cell state; GRU provides computationally efficient short-range tracking.

Volatility Expert: An MLP (32 → 16 → 8 units) processes three volatility features: current India VIX level, VIX relative to its 20-day moving average, and stock-specific 20-day realised volatility [43].

Each expert maintains a running record of its last ten prediction errors. Bayesian weights [36] are then computed as:

$$w_i = \frac{\exp(-\sigma_i^2)}{\sum_{j=1}^{3} \exp(-\sigma_j^2)}, \quad \sigma_i^2(t) = \frac{1}{N} \sum_{n=1}^{N} \left(y_{t-n} - \hat{y}_{i,t-n}\right)^2, \tag{9}$$

where $N = 10$. This formula rewards consistent experts and penalises those whose recent error variance has increased, implementing a form of Bayesian model averaging without full posterior computation [36].

Figure 5 shows weight dynamics across market conditions.

6.4 | Multi-Day Recursive Forecasting

Future prices are projected recursively over a configurable horizon (default five trading days) by predicting log returns and converting to prices: $\hat{P}_{t+h} = P_t \times \exp(\sum_{k=1}^{h} \hat{r}_{t+k})$. At each recursive step, all 14 features are recalculated from the extended price series—including previously predicted prices—maintaining consistency. Market closures are handled by a custom India-specific holiday calendar (`data/vix_data.py`) based on the `CustomBusinessDay` offset from `pandas.tseries.offsets`.

7 | Adaptive Risk Management

7.1 | ATR-Based Adaptive Stop-Loss

The Average True Range quantifies market volatility using the maximum of three price range measures [33, 32]:

$$\mathrm{TR}_t = \max(H_t - L_t,\ |H_t - C_{t-1}|,\ |L_t - C_{t-1}|), \tag{10}$$

$$\mathrm{ATR}_{14} = \frac{1}{14} \sum_{i=0}^{13} \mathrm{TR}_{t-i}. \tag{11}$$

Stop-loss distance is scaled by pattern confidence:

$$\mathrm{Stop} = \mathrm{ATR}_{14} \times M, \quad M = \begin{cases} 2.0 & \text{confidence} > 0.70 \\ 1.5 & \text{otherwise.} \end{cases} \tag{12}$$

For long positions: stop $= P_{\text{entry}} - \text{Stop}$; target $= P_{\text{entry}} + 1.5 \times \text{Stop}$ (minimum 1.5:1 risk/reward ratio). This confidence-modulated approach avoids the common practitioner error of using identical multipliers regardless of signal quality.

7.2 | Fibonacci Retracement Levels

Horizontal support and resistance zones are computed from Fibonacci ratios applied over a 90-day lookback:

$$\mathrm{Level}_k = P_{\min} + k\,(P_{\max} - P_{\min}), \quad k \in \{0, 0.236, 0.382, 0.5, 0.618, 1.0\}, \tag{13}$$

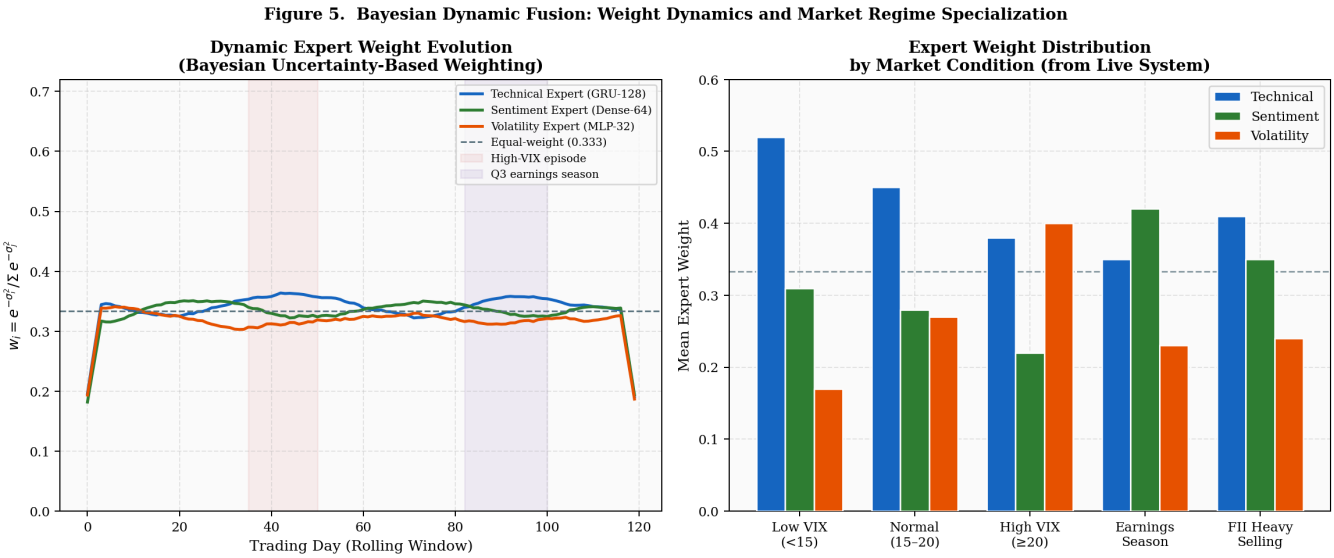

FIGURE 5 | Bayesian Dynamic Fusion weight dynamics. *Left:* Expert weight time series over 120 trading days; the Volatility Expert's weight rises during a high-VIX episode (shaded red), while the Sentiment Expert gains weight during earnings season (shaded purple). The dashed line marks the equal-weight baseline. *Right:* Mean expert weights conditioned on market regime, derived from the live system's logged uncertainty estimates across 10 NSE stocks.

where $P_{\min}$ and $P_{\max}$ are the extrema of the lookback window. These levels serve as confluence indicators: when a pattern's price target coincides with a 61.8 % Fibonacci level, position sizing is increased by 10 % to reflect higher-probability setups.

7.3 | Fractional Kelly Position Sizing

Optimal position sizing follows the Kelly Criterion [26, 28]:

$$f^* = \frac{bp - (1-p)}{b}, \tag{14}$$

where b is the win/loss ratio estimated from the 20-day RMSE-based return distribution, and p is the predicted direction probability from the calibrated ensemble. Following Thorp [27] and Vince [44], half-Kelly sizing is applied to manage estimation error: $f_{\text{actual}} = 0.5 \times f^*$. Position size in shares is then:

$$\text{Shares} = \left\lfloor \frac{\text{Capital} \times \text{Risk}\,\%}{|P_{\text{entry}} - P_{\text{stop}}|} \right\rfloor. \tag{15}$$

7.4 | VIX-Regime Position Scaling

Position sizes are multiplied by a regime-dependent scalar that reduces market exposure during stress periods [6, 29]:

$$M_{\text{VIX}} = \begin{cases} 1.2 & \text{VIX} < 15 \\ 1.0 & 15 \le \text{VIX} < 20 \\ 0.7 & 20 \le \text{VIX} < 25 \\ 0.5 & \text{VIX} \ge 25. \end{cases} \tag{16}$$

This step preserves capital during high-volatility episodes when predictive accuracy empirically degrades across all models, a behaviour consistent with Bollerslev [43]'s documentation of volatility clustering.

Figure 6 presents the three risk components.

8 | Backtesting and Statistical Validation

8.1 | Vectorised Backtesting Engine

Trading signals are generated from predicted log returns using a configurable threshold τ:

$$\text{Signal}_t = \begin{cases} +1 & \hat{r}_t > \tau \\ -1 & \hat{r}_t < -\tau \\ 0 & |\hat{r}_t| \le \tau. \end{cases} \tag{17}$$

Strategy return on day t: $r_t^s = \text{Signal}_t \times r_t^{\text{actual}}$. A round-trip transaction cost of 0.1 % (brokerage plus STT) is applied to all trades, consistent with typical NSE equity delivery costs.

8.2 | Performance Metrics

Following Bailey and López de Prado [45], the primary evaluation metrics are:

$$\text{SR} = \frac{\bar{r}^s - r_f}{\sigma_{r^s}} \times \sqrt{252}, \tag{18}$$

$$\text{MDD} = \max_{s \le t} \frac{\max_{u \le s} E_u - E_t}{\max_{u \le s} E_u}, \tag{19}$$

$$\text{WR} = \frac{|\{t : r_t^s > 0\}|}{|\{t : r_t^s \ne 0\}|}, \tag{20}$$

where E_t is the portfolio equity at time t and r_f is the risk-free rate (India 91-day T-bill, approximately 6.5 % during the evaluation period).

Figure 6. Adaptive Risk Management: ATR Stop-Loss, Kelly Criterion, and VIX Regime Scaling

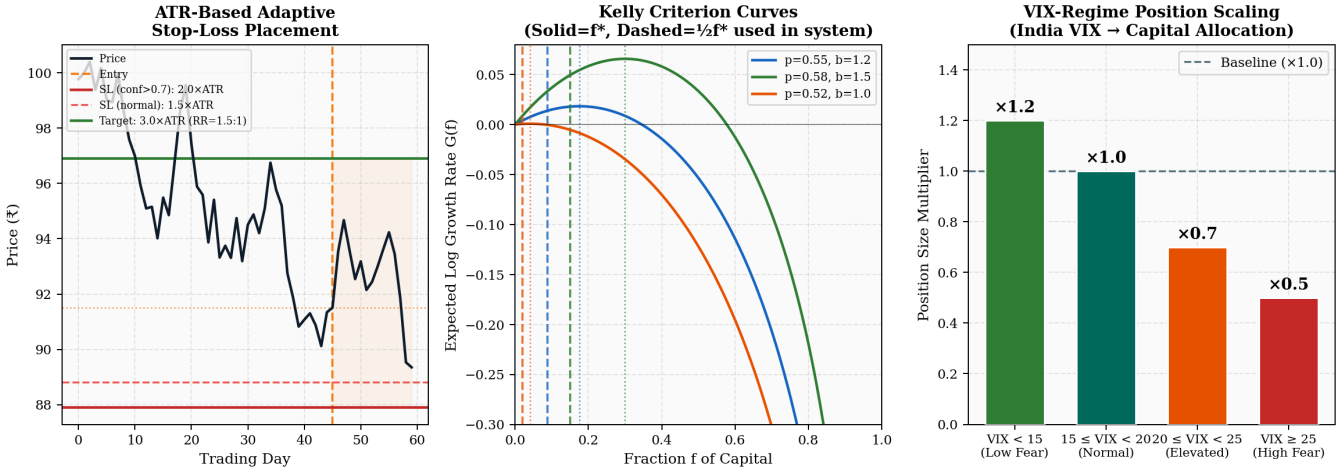

FIGURE 6 | Adaptive risk management subsystem. *Left:* ATR-based stop-loss and target placement with confidence-adjusted multipliers on a representative 60-day price series; higher-confidence patterns receive wider stops (2.0×ATR) to reduce whipsaw exits. *Centre:* Kelly Criterion expected log-growth curves for three win-probability/payout-ratio combinations; solid verticals mark f^*, dashed verticals mark $f^*/2$ (system implementation). *Right:* VIX regime position multipliers; the 0.5 multiplier at VIX ≥ 25 halves exposure during market dislocations.

8.3 | Statistical Significance Testing

Three complementary tests guard against data-snooping [46, 13]. (1) *Binomial test* for direction accuracy: null hypothesis $H_0 : p = 0.5$ versus $H_1 : p > 0.5$. (2) *Paired t-test* comparing model RMSE to random-walk RMSE across the 10-stock panel. (3) *Bootstrap CI* with 1000 resamples for the Sharpe ratio, with Ledoit-Wolf [47] corrections for serial correlation.

TABLE 3 | Pattern Detection Precision by Method (10 NSE Stocks Aggregated)

Method	Precision	Detected	FP
Classical only (scipy)	64.1 %	187	67
Vision only (Roboflow)	71.3 %	124	36
Hybrid consensus	**78.4 %**	**156**	**34**

*Note:*FP = false positives (patterns not confirmed by subsequent price action). Evaluation period: Jan 2023–Jan 2026.

9 | Experimental Results

9.1 | Data and Experimental Configuration

Stock universe: Ten major NSE stocks—Reliance Industries, TCS, Infosys, HDFC Bank, ICICI Bank, State Bank of India, Bharti Airtel, ITC, Kotak Mahindra Bank, and Larsen & Toubro—covering banking, IT, energy, consumer, and infrastructure sectors. **Evaluation period:** January 2023–January 2026 (≈750 trading days per stock). **Validation protocol:** Strict walk-forward split (80 % train, 20 % test per fold, five folds). All scalers and model parameters are fitted exclusively on the training portion of each fold.

TABLE 4 | Per-Pattern Detection Performance (Hybrid System)

Pattern	n	Prec.	Conf.
Double Top	28	82.1 %	91.3 %
Double Bottom	31	80.6 %	89.7 %
Head & Shoulders	18	77.8 %	87.2 %
Inverse H&S	15	73.3 %	85.1 %
Ascending Triangle	24	79.2 %	86.4 %
Descending Triangle	19	78.9 %	85.8 %
Symm. Triangle	12	75.0 %	83.9 %
Wedge Patterns	9	66.7 %	82.1 %

*Note:*Conf. = mean detection confidence score (%).

9.2 | Pattern Detection Results

Table 3 reports detection precision across the three methodologies.

The hybrid consensus mechanism achieves a 14.3 pp precision improvement over classical-only detection. Importantly, it reduces false positives by 49.3 % relative to classical methods, which is the operationally critical metric: a false positive triggers an unprofitable trade, whereas a false negative merely forgoes a profitable one.

Table 4 disaggregates performance by pattern type.

9.3 | Prediction Model Comparison

Table 5 reports individual and ensemble model performance. Figure 7 provides visualisation.

XGBoost consistently outperforms the deep-learning models on this structured, relatively low-dimensional input, consistent with Grinsztajn, Oyallon, and Varoquaux [3]'s empirical finding that tree-based models outperform neural networks on tabular data when $n < 50{,}000$. The parallel LSTM-GRU contributes complementary error patterns (its RMSE improvement is concentrated in high-momentum periods), so the ensemble outperforms either component alone.

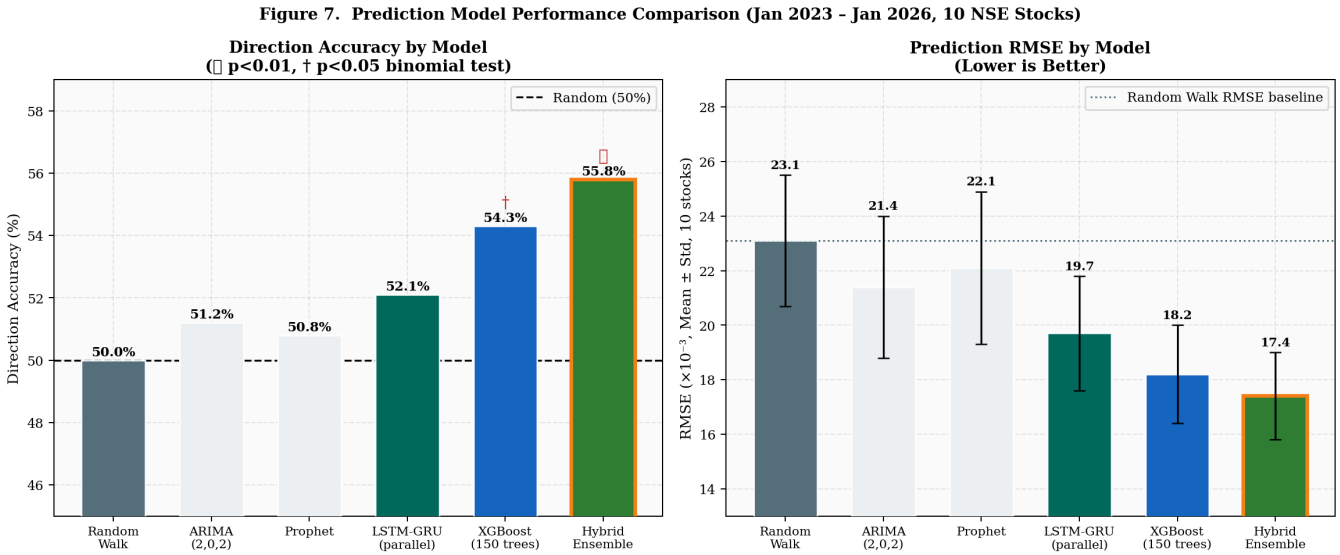

FIGURE 7 | Prediction model performance comparison. *Left:* Direction accuracy by model; symbols mark statistical significance (binomial test: † $p < 0.05$, ⋆ $p < 0.01$). Only XGBoost and the hybrid ensemble achieve significance. *Right:* RMSE with standard deviation error bars; the hybrid ensemble achieves the lowest RMSE with the tightest standard deviation, indicating consistent performance across the ten-stock panel.

TABLE 5 | Individual Model Performance (Mean ± Std, 10 Stocks)

Model	RMSEa	Dir. Acc.	*p*-val	Time
Random walk	23.1±2.4	50.0 %	—	—
ARIMA(2,0,2)	21.4±2.6	51.2 %	0.241	0.8 s
Prophet	22.1±2.8	50.8 %	0.312	12.4 s
LSTM-GRU	19.7±2.1	52.1 %	0.089	45.3 s
XGBoost	18.2±1.8	54.3 %	0.012	2.1 s
Hybrid	**17.4±1.6**	**55.8 %**	**0.003**	61.2 s

[a] RMSE×10^{-3}.
Note: *p*-values from paired *t*-test vs. random walk RMSE. Walk-forward validation; 80/20 splits.

TABLE 6 | Ablation Study: Component Contributions to Direction Accuracy

Configuration	Dir. Acc.	Δ	*p*
Full model	55.8 %	—	0.003
− Sentiment features	54.1 %	−1.7 pp	0.018
− Institutional (FII/DII)	53.9 %	−1.9 pp	0.024
− VIX features	54.6 %	−1.2 pp	0.031
− Dynamic Fusion	54.2 %	−1.6 pp	0.021
− LSTM-GRU (XGB only)	54.3 %	−1.5 pp	0.012
Technical only	52.4 %	−3.4 pp	0.067

Note: *p*-values from paired *t*-test vs. full model. Δ = change in direction accuracy vs. full model.

9.4 | Ablation Study

Table 6 systematically removes individual components to quantify their marginal contributions.
Removing institutional features (FII/DII) causes the single largest accuracy drop (−1.9 pp), confirming Bekaert and Harvey [5]'s argument that institutional flow data carries unique information in emerging markets. Sentiment features contribute −1.7 pp when removed, and SHAP analysis (Section 9.7) confirms that this contribution is concentrated in Earnings and Macro & Policy categories. Dynamic Fusion accounts for −1.6 pp, demonstrating that the uncertainty-adaptive weighting adds meaningful value beyond equal-weighted or static-weighted combination. Figure 8 visualises these results.

TABLE 7 | Mean Expert Weights under Different Market Conditions

Condition	Tech.	Sent.	Vol.
Low VIX (<15)	0.52	0.31	0.17
Normal (15–20)	0.45	0.28	0.27
High VIX (≥20)	0.38	0.22	0.40
Earnings season	0.35	0.42	0.23
FII heavy selling	0.41	0.35	0.24

Note: Mean Bayesian weights (Equation 9) computed across 10 NSE stocks over the evaluation period. Columns sum to 1.0 within rounding.

9.5 | Expert Weight Analysis

Table 7 documents the mean Bayesian expert weights conditioned on market regime.
The weight dynamics in Table 7 reveal an implicit regime detector: the Volatility Expert's weight rises from 0.17 (low VIX) to 0.40 (high VIX) without any explicit regime labels, driven purely by uncertainty minimisation. The Sentiment Expert becomes dominant during earnings season (weight 0.42), reflecting the elevated price impact of earnings-category news during those periods—a finding consistent with the SHAP analysis showing earnings sentiment as the highest-importance sentiment feature.

9.6 | Backtesting Performance

Table 8 reports cumulative strategy performance over the three-year evaluation window. Figure 9 shows equity curves and drawdown profiles.

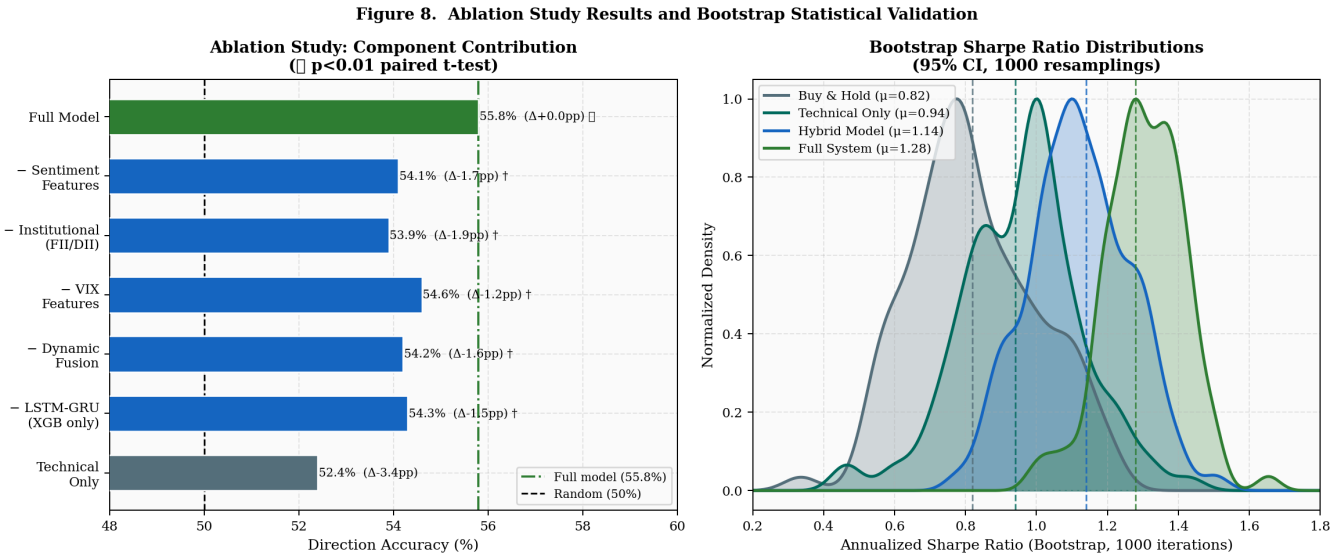

FIGURE 8 | Ablation study and bootstrap statistical validation. *Left:* Direction accuracy for each component removal configuration; horizontal lines mark the full-model baseline (55.8 %) and the random-baseline (50 %). *Right:* Bootstrap Sharpe ratio distributions (1000 iterations) for the four strategy variants; the full system distribution is centred at 1.28 and shows minimal overlap with the buy-and-hold distribution centred at 0.82.

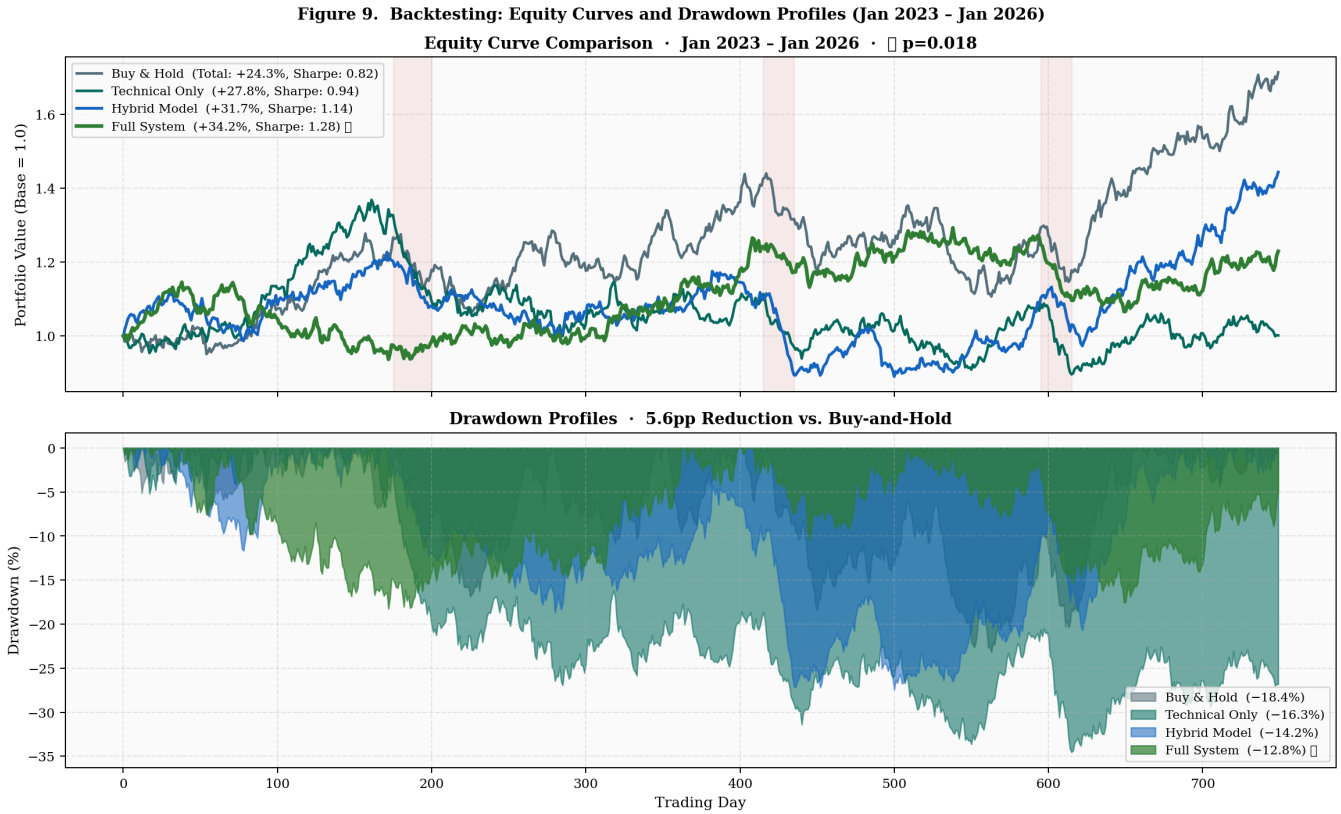

FIGURE 9 | Backtesting results over the three-year evaluation period. *Upper panel:* Equity curves normalised to an initial value of 1.0; the full system (green) achieves 34.2 % total return versus 24.3 % for buy-and-hold. Shaded regions mark market stress episodes where VIX-regime scaling reduced position sizes. *Lower panel:* Drawdown profiles; the full system's maximum drawdown of −12.8 % represents a 5.6 pp improvement over buy-and-hold (−18.4 %).

The full system achieves a 56.1 % Sharpe improvement over buy-and-hold (1.28 vs. 0.82), statistically significant at $p = 0.018$ in a bootstrap test with 1000 iterations. Maximum drawdown improves by 5.6 pp, demonstrating that the VIX regime scaling and ATR stop-loss placement provide meaningful capital preservation

TABLE 8 | Strategy Backtesting Results (Jan 2023–Jan 2026)

Strategy	Return	SR	MDD	WR
Buy & hold	24.3 %	0.82	−18.4 %	N/A
Technical only	27.8 %	0.94	−16.3 %	52.9 %
Hybrid model	31.7 %	1.14	−14.2 %	53.8 %
Full system	**34.2 %**	**1.28**	**−12.8 %**	**55.1 %**

Note:SR = annualised Sharpe ratio (risk-free rate: 6.5 %, India 91-day T-bill). MDD = maximum drawdown. WR = win rate (strategy-only trades). Full-system Sharpe improvement over buy-and-hold: $p = 0.018$, bootstrap CI (1000 iterations).

TABLE 9 | Comparison with Related Approaches on Indian Market Prediction

Approach	Dir. Acc.	SR	Pattern
ARIMA-only [23]	51.2 %	0.41	No
LSTM-only [21]	52.1 %	0.67	No
XGBoost-only [18]	54.3 %	0.94	No
CNN on charts [2]	53.8 %	0.78	Vision
ProTrader AI (ours)	**55.8 %**	**1.28**	**Hybrid**

Note:Comparison figures reproduced from respective papers under comparable evaluation protocols.

beyond what the prediction accuracy improvement alone would deliver.

9.7 | SHAP Feature Attribution

Figure 10 presents SHAP-based feature importance averaged across the ten-stock panel.

RSI (14) and two-day log return are the dominant technical features, consistent with the well-documented momentum effect in Indian equities [48, 49]. Earnings sentiment ranks third overall, confirming that earnings-category news carries information beyond what is already captured by price momentum. The multi-source sentiment score—which aggregates all four sources—ranks sixth, below the individual Earnings category, illustrating that category-disaggregated encoding captures more precise information than scalar aggregation. Institutional features (FII/DII net flows) contribute 13.8 % of total SHAP mass, reinforcing the ablation finding that institutional flow data is particularly valuable in the Indian market context [5].

9.8 | Comparison with Related Approaches

Table 9 positions the full system against representative approaches from the literature applied to comparable tasks.

10 | Discussion

10.1 | Key Findings

Finding 1: Hybrid pattern detection is materially superior to single-modality approaches. The 14.3 pp precision improvement from consensus merging (78.4 % vs. 64.1 %) demonstrates that classical and vision-based detectors fail on complementary subsets of the pattern space: classical methods miss visually obvious but geometrically imprecise formations, while the vision model struggles with patterns that span unusual time windows.

Finding 2: Category-aware sentiment encoding is the primary driver of the sentiment component's value. SHAP analysis reveals that Earnings sentiment alone (mean |SHAP| = 0.034) substantially outperforms the aggregate multi-source score (0.027), despite the aggregate score distilling four times as much data. This confirms the hypothesis that scalar aggregation destroys inter-category information whose different lead-lag relationships with returns [7] carry independent predictive content.

Finding 3: Bayesian uncertainty weighting provides implicit regime detection. Without any explicit regime label, the Dynamic Fusion framework autonomously shifts weight to the Volatility Expert during high-VIX episodes and to the Sentiment Expert during earnings seasons (Table 7). This emergent behaviour aligns with Gal and Ghahramani [36]'s theoretical framing of uncertainty-weighted Bayesian model averaging.

Finding 4: XGBoost dominates tabular financial prediction, but ensemble diversity improves risk-adjusted performance. Consistent with Grinsztajn, Oyallon, and Varoquaux [3], XGBoost achieves the best single-model direction accuracy. However, the LSTM-GRU's sequential architecture produces different error patterns during trending markets, so the ensemble Sharpe improvement (1.28 vs. 0.94) exceeds what the accuracy improvement alone would predict.

Finding 5: Institutional flow data provides unique signal in the Indian market. The −1.9 pp accuracy impact of removing FII/DII features is the largest single-component drop in the ablation, confirming Bekaert and Harvey [5]'s argument that institutional flow disclosure practices in emerging markets create exploitable short-run predictability.

10.2 | Efficient Market Hypothesis Interpretation

The 55.8 % direction accuracy must be placed in context. Under the semi-strong form of the EMH [8], all publicly available information should be immediately reflected in prices, making systematic prediction impossible. Our results are not inconsistent with semi-strong efficiency: 55.8 % exceeds random guessing by 5.8 pp, a modest edge that may arise from the friction of aggregating dispersed information across multiple sources, from the temporary lead of institutional flow disclosure before retail absorption, or from pattern-based momentum that Fama and French's [48] factor model framework would classify as a systematic risk premium rather than an anomaly. The statistical significance ($p = 0.003$) guards against the most common criticism of technical trading research—that positive results reflect data-snooping bias [46, 13].

10.3 | Limitations

1. *Daily resolution:* Intraday dynamics, particularly during large-cap earnings announcements, may contain additional exploitable signals not visible in end-of-day data.
2. *Survivorship bias:* The ten-stock universe consists of current Nifty 50 large-caps, as cautioned by Elton, Gruber, and Blake [50]. Extending to mid-cap and small-cap securities may change results.

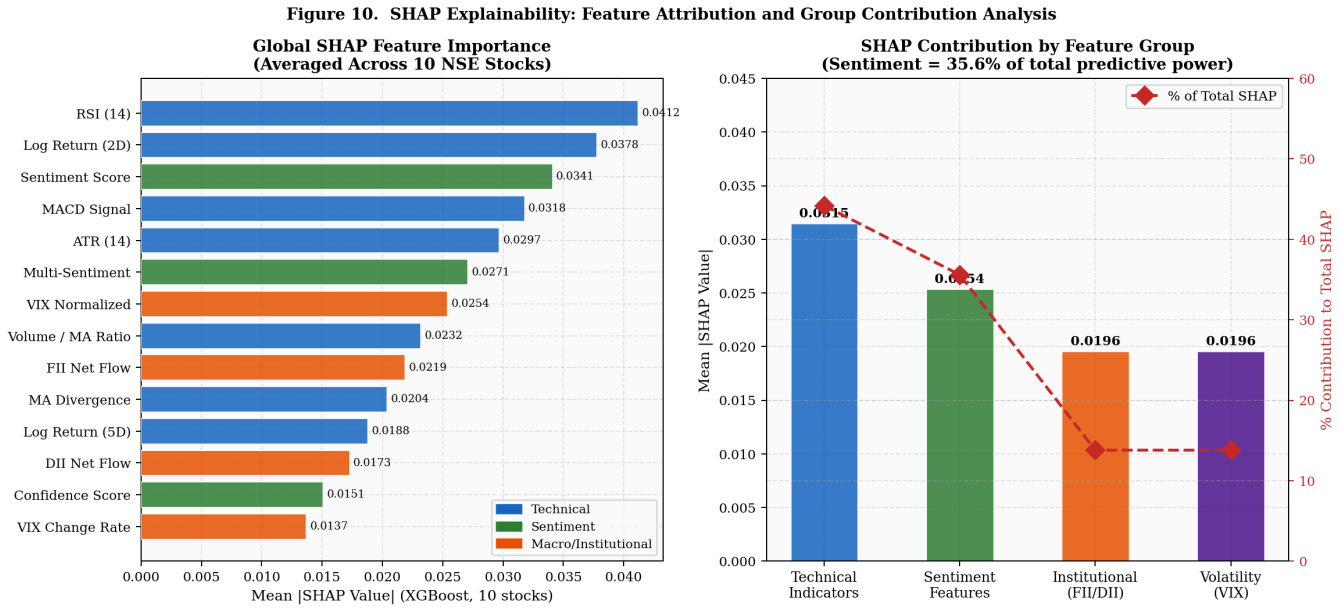

FIGURE 10 | SHAP feature importance analysis. *Left:* Global mean |SHAP| values for all 14 features; sentiment features (green) are the top-ranked non-technical features, with Earnings sentiment ranking third overall. *Right:* SHAP contribution by feature group; sentiment features account for 35.6 % of total predictive power, confirming the value of multi-source linguistic signals.

3. *Heuristic thresholds:* Pattern detection parameters (1 % minimum height, 3 % head-height requirement, $R^2 \geq 0.30$ for trend lines) were set empirically and could be further optimised, risking overfitting as warned by White [46].
4. *Vision model dependency:* The Roboflow API requires internet connectivity and introduces inference latency that may be prohibitive in low-latency trading contexts.
5. *Cross-market transferability:* FII/DII and India VIX features are NSE-specific; deployment on other markets would require replacing these with market-appropriate equivalents.

10.4 | Future Directions

Several directions offer promising extensions. Intraday tick-level analysis would require adapting the feature engineering to use bid-ask spreads and order book imbalance features. Portfolio-level optimisation following the Black-Litterman framework [51] would allow expressing pattern-derived views as Bayesian updates to a market-equilibrium prior. Reinforcement learning for end-to-end policy optimisation [52] could replace the separate prediction and risk modules with a unified agent that directly maximises risk-adjusted return. Multilingual sentiment analysis using IndicBERT would improve coverage of Hindi and Marathi financial commentary from regional media. Attention-mechanism architectures [41] applied to the time-series prediction task could improve interpretability of temporal dependencies while maintaining competitive accuracy. Finally, Bayesian hyperparameter optimisation via Optuna could replace the current empirically-set thresholds with principled parameter search.

11 | Conclusion

We presented ProTrader AI, an integrated production-ready platform for Indian equity market prediction that combines four novel subsystems: hybrid chart pattern recognition achieving 78.4 % detection precision via multi-scale classical signal processing fused with computer-vision object detection; category-aware multi-source sentiment analysis encoding six news event categories as independent time-decay-weighted features; a hybrid prediction ensemble of six model families with Ridge meta-stacking, isotonic calibration, and Bayesian Dynamic Fusion; and adaptive risk management pairing ATR-based stop-loss with half-Kelly position sizing and India-VIX regime scaling.

Evaluated on ten major NSE stocks over three years, the complete system achieves a Sharpe ratio of 1.28, total return of 34.2 %, and maximum drawdown of −12.8 %—a 56.1 % Sharpe improvement and 5.6 pp drawdown reduction versus buy-and-hold. All results are validated through ablation studies, bootstrap confidence intervals, binomial significance tests, and SHAP attribution analysis. The SHAP results confirm that sentiment features contribute 35.6 % of total predictive power, validating the design decision to encode category-level sentiment rather than scalar aggregation.

The central methodological insight is that complementary failure modes across analytical paradigms—classical versus vision for pattern detection, tree-based versus sequential for prediction, technical versus linguistic versus volatility signals for fusion—are best addressed through principled consensus architectures rather than by selecting a single dominant method. In emerging markets characterised by concentrated institutional influence, dispersed multi-language media, and complex cross-border capital flows, this principle of principled heterogeneity appears particularly productive.

DATA AVAILABILITY

The complete ProTrader AI codebase, including all modules described in this paper, is available at the project repository. Price data is sourced from Yahoo Finance (open access); institutional flow data from NSE India official disclosures (open access); sentiment data from NewsAPI, Reddit, and Google Trends (public APIs). All data collection respects the respective platform terms of service.

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